

Title

Matrix Operations Tool using NumPy

Objective

To perform various matrix operations through a Python-based interactive tool.

Operations Implemented

- Addition
- Subtraction
- Multiplication
- Transpose
- Determinant

Tools Used

- Python
- NumPy

Results

===== MATRIX OPERATIONS TOOL =====

1. Addition
2. Subtraction
3. Multiplication
4. Transpose
5. Determinant
6. Exit

Enter your choice (1-6): 5

Enter size of square matrix: 2

Enter Matrix:

1 2

3 4

Determinant:

-2.0000000000000004

===== MATRIX OPERATIONS TOOL =====

1. Addition
2. Subtraction
3. Multiplication
4. Transpose
5. Determinant
6. Exit

Enter your choice (1-6): 4

Enter number of rows: 2

Enter number of columns: 3

Enter Matrix:

1 2 3

4 5 6

Transpose:

[[1. 4.]

[2. 5.]

[3. 6.]]

===== MATRIX OPERATIONS TOOL =====

1. Addition
2. Subtraction
3. Multiplication
4. Transpose
5. Determinant
6. Exit

Enter your choice (1-6): 3

Enter number of rows: 2

Enter number of columns: 2

Enter Matrix A:

1 2

3 4

Enter Matrix B:

5 6

7 8

Result:

[[19. 22.]

[43. 50.]]

===== MATRIX OPERATIONS TOOL =====

1. Addition
2. Subtraction
3. Multiplication
4. Transpose
5. Determinant
6. Exit

Enter your choice (1-6): 2

Enter number of rows: 2

Enter number of columns: 2

Enter Matrix A:

8 7

6 5

Enter Matrix B:

1 2

3 4

Result:

[[7. 5.]

[3. 1.]]

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===== MATRIX OPERATIONS TOOL =====
1. Addition
2. Subtraction
3. Multiplication
4. Transpose
5. Determinant
6. Exit
Enter your choice (1-6): 1
Enter number of rows: 2
Enter number of columns: 2

Enter Matrix A:
1 2
3 4

Enter Matrix B:
5 6
7 8

Result:
[[ 6.  8.]
 [10. 12.]]
```

Conclusion

The application successfully performs matrix operations using NumPy.